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ABSTRACT

The project “Detection and Recommendations in Peanut Farming” focuses on detecting pests, diseases, and crop damage in peanut plants using machine learning and deep learning algorithms. It analyzes various plant parts such as stems, pods, roots, leaves, and insects to provide complete crop health assessment. The system employs the YOLOv8 deep learning model for accurate image-based detection of pests and visible symptoms, while a Random Forest algorithm is used for classification and prediction of disease severity. The approach combines visual analysis and predictive modeling to identify affected areas and recommend suitable pesticides. This intelligent system helps farmers detect issues early, minimize crop loss, and make data-driven decisions to improve productivity and sustainability in peanut cultivation.

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LIST OF ACRONYMS

Acronym	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
FastAPI	Fast Application Programming Interface
GPU	Graphics Processing Unit
HTML	HyperText Markup Language
JSON	JavaScript Object Notation
mAP	Mean Average Precision
ML	Machine Learning
OpenCV	Open Source Computer Vision Library
RF	Random Forest
UI	User Interface
UML	Unified Modeling Language
YOLO	You Only Look Once
YOLOv8	You Only Look Once version 8